

Statistical Analysis of Monthly Expences of Hostellers

A project report submitted in partial fulfilment of
requirement for the degree of B.Sc.
(Statistics/Actuarial) with specialization in
Mathematical Sciences

Submitted by

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Submitted to

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Jalgaon
(2025-2026)

CERTIFICATE

This is to certify that **Mr. Hitesh Rameshwar Jadhav** and **Mr. Manish Narendra Patil** are the students of B.Sc. II Mathematical Sciences, at Kavayitri Bahinabai Chaudhari North Maharashtra University, Jalgaon. They have successfully completed their project work entitled “ Statistical Analysis of Monthly Expenses of Hostellers” based on the survey conducted under my guidance and supervision during the academic year 2025- 2026.

Date: 19 Nov 2025

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ABSTRACT

This study investigates the ‘Monthly Expenses of Hostellers ’ for hostelers. The research aims to explore the relationship between monthly expenses and hostelers. The study uses a quantitative research design and collects data through an online survey. The data is analysed using statistical tools such as descriptive statistics and statistical charts and figures. The study finds the average monthly expenses of hostelers based on gender. The data analysis shows whether is there any association between gender and expenses.

ACKNOWLEDGEMENTS

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Mr Hitesh. R. Jadhav

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Chapter 1 :- Introduction

1.1 Introduction :

This field project aims to investigate and Analyze the Monthly Expenditure patterns of hostel residents, focusing on various categories of expenses they incur during their stay. Hostelers often face unique financial challenges as they live away from home, balancing both basic needs and lifestyle choices. Understanding these expenses is crucial for better financial planning and management, not only for hostelers themselves but also for hostel administrators and financial planners.

The project will collect data through surveys, and observations, with a focus on essential categories like food, accommodation, utilities, transportation, and entertainment. By categorizing these expenses, the study will shed light on the economic pressures faced by hostel residents, highlighting how they prioritize their spending and manage their limited budgets.

1.2 Objectives :

- To Analyze the Monthly Expenditure Pattern of Hostelers
- To Identify the Most Common Sources of Income for Hostelers
- To Assess the Cost of Living in Different Types of Hostels and To Evaluate the Impact of Hostel Expenses on Hostelers Financial condition
- To Understand the Variations in Expenses Among Different Groups of Hostelers
- To Determine the Effect of Hostel Expenses on Students' Lifestyle Choices

1.3 Scope :

1.3.1 Project Objective :

- **Purpose:** To understand the breakdown of the monthly expenses incurred by students living in hostels.
- **Scope:** The project will focus on hostelers from different hostels and regions, comparing their expenses in categories like food, rent, utilities, etc.

1.4 Limitations of Study :

1.4.1 Sample Size and Representativeness: The sample may not fully represent the entire population of hostelers, especially if it's limited to one region or a small number of hostels.

1.4.2 Self-Reported Data: Relying on self-reported information from hostelers might lead to inaccuracies or biases in how expenses are remembered or reported.

1.4.3 Variability of Hostel Expenses: Different hostels may have vastly different costs, making it difficult to draw generalized conclusions that apply across all hostels.

Chapter 2 :– Data Collection/Exploration

The data which we collected for our project is **purely primary data**. So, from lots of questions, opinions , suggestions, and guidance of our supervisor we decided to make a questionnaire using google form. In the way of collecting data, we have to decide one thing that is what sample size should be kept to make analysis more convenient and reliable so we decided to stop our collection at **100 responses**.

Making of this form has a so many challenges such as what questions we have to ask, we decided to ask the following questions :

2.1 Questionnaire :

- 1) What is your age?
 - I. 18-20
 - II. 21-23
 - III. 24-26
- 2) Gender
 - I. Male
 - II. Female
- 3) How long have you been living in hostel ? (years)
 - I. 1 yr
 - II. 2 yr
 - III. 3 yr
 - IV. 4 yr
 - V. 5 yr
- 4) From where do you belong?
 - I. Rural
 - II. Urban
- 5) Current education status?
 - I. U.G first year
 - II. U.G second year
 - III. U.G third year
 - IV. U.G fourth year
 - V. P.G first year
 - VI. P.G second year
- 6) What type of room are you staying in?
 - I. Single
 - II. Double
 - III. Triple
- 7) How much do you spend on travelling per month?
 - I. 0-250
 - II. 250-500
 - III. Above 500

8) How much do you spend on food ? (including mess and other per month)

- I. 2000-3000
- II. 3000-4000
- III. Above 4000

9) How much do you spend on entertainment per month?

- I. 0-500
- II. 500-1000
- III. Above 1000

10) How much do you spend on study material per month?

- I. 0-100
- II. 100-200
- III. Above 200

11) Do you have any additional source of income?

- I. Yes
- II. No

12) How much do you spend for rent per month?

- I. 4500
- II. 5000
- III. 5500

13) How much do you spend on mobile recharge per month?

- I. 200-300
- II. 300-500
- III. Above 500

14) How much do you spend on personal or health care per month?

- I. 0-500
- II. 600-1000
- III. Above 1000

15) Do hostel need to provide WiFi?

- I. Yes
- II. No

16) Father occupation?

- I. Farmer
- II. Teacher
- III. Pvt. Job
- IV. Gov. Job
- V. Businessman
- VI. Other

17) Mode of payment?

- I. Online
- II. Cash

18) Averagely how much do you spend per month?

- I. Upto 2000
- II. 2000-4000
- III. Above 4000

2.2 Cleaning of Data :

Thereafter collecting responses of all of this data we have to clean this data. For this we check the data for duplicates and remove them. Duplicates can skew our data and make analysis more difficult. Then we remove outliers to prevent them from skewing our analysis.

2.3 Tabulation of the Data :

After collecting all the data, we make table of our collected data and sorted in rows and columns of different kinds, different characteristics and as need of our project. We have done our tabulation using Ms-Excel software in MS-office. Also, we use Minitab for chi-square test and determine the associations . Tabulation can be used for a variety of purpose, such as analysing survey results, summarizing demographic information. Here we use simple tabulation and coloured tabulation for identifying the data variation and complexity. Given below are some pictures of our data set collected and well arranged in tabulation format:

2.4 Data Collection

2.4.1 Target Group: Students currently residing in hostels.

2.4.2 Methods:

- Surveys/Questionnaires to collect data on monthly expenses.
- Interviews with a sample of hostelers.
- Observation or informal interactions.

2.4.3 Sample Size: Minimum of 100 hostelers for reliable data collection.

2.5 Expense Categories

The project will focus on identifying and calculating the average expenditure in the following categories:

- Rent: Monthly fee paid for hostel accommodation.
- Food: Costs for meals (whether provided by the hostel or eaten outside).
- Transportation: Costs related to travel to and from the hostel.
- Study Materials: Books, stationery, etc.
- Entertainment: Spending on social activities, cinema, etc.
- Miscellaneous: Other costs, including personal expenses or emergencies.

2.6 Data Analysis

- Calculate the average monthly expense per student for each category.
- Identify any variations in spending patterns based on geographical location (urban vs. rural).

2.7 Resources Needed

- Survey tools (digital forms or paper-based surveys).
- Analytical tools (spreadsheet software for data analysis).

Chapter 3 :- Methods and Statistical tools

There are various methods we used in our project for collecting data, analysing it, making tables, interpreting it, and many more. Here are the following:

3.1 Survey Method :

A survey is a research method that involves asking a set of structured questions to a sample of individuals in order to gather information about a specific topic or issue. Surveys can be conducted using various methods such as online surveys or face to-face interviews. Here we use google forms to collect information from hostelers google is the new modern way to collect data virtually without using paper.

3.2 Data Scrubbing Method :

It is also known as data cleansing, which is the process of identifying and correcting errors, inconsistencies, and inaccuracies in the data. It involves detecting and handling incomplete, duplicate, incorrect, or irrelevant data to ensure that the data is accurate, consistent, and reliable for analysis. The goal of data cleaning is to improve the quality of data, remove errors, and increase the accuracy of analysis results.

3.3 Data Analysis :

we analysed the cleaned data using statistical tools to determine whether hostel expenses is associated with anything or not . We use various statistical tools for this such as chi-square test , descriptive statistics, statistical charts, frequency tables.

3.4 Data Visualization :

Data visualization is the representation of data and information in visual formats such as charts, graphs, and maps. The purpose of data visualization is to provide a clear and concise understanding of complex data and patterns by presenting it in a graphical format that is easy to interpret and analyse. Data visualization is an important tool for data analysis, as it can help to identify trends, patterns, and relationships in data that might not be immediately apparent through numerical analysis alone. Some common types of data visualization include horizontal bar charts, doughnut charts, different pie charts, among others.

3.5 Software's :

We use various software for statistical analysis, such as **Microsoft Excel, Minitab and Microsoft Word**. The data we collected in the Google Form is then converted into an Excel file and R script for analysis purposes, and we make our interpretation from this file.



Microsoft excel



MiniTab



Microsoft word

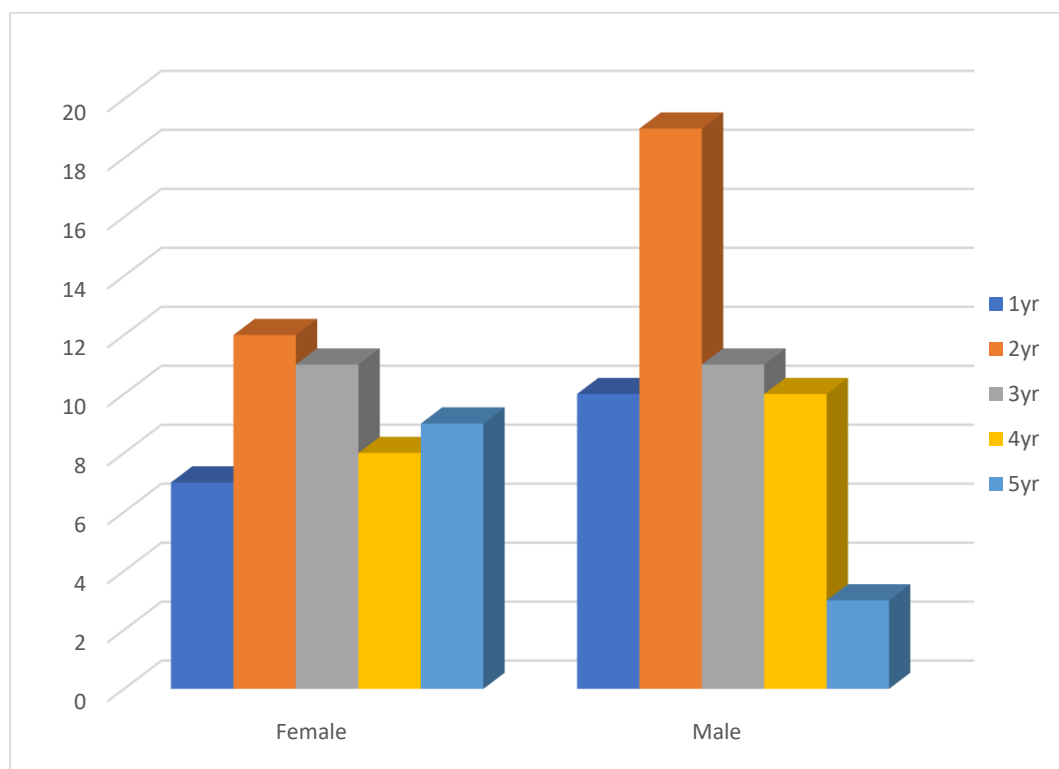
Chapter 4 :- Exploratory Data Analysis

4.1 Graphical visualization:

Table 4.1: Frequency table of gender and years lived in hostel.

Count of Gender	How long have you been living in hostel					
Gender	1	2	3	4	5	Grand Total
Female	7	12	11	8	9	47
Male	10	19	11	10	3	53
Grand Total	17	31	22	18	12	100

Fig 4.1: Multiple bar diagram of table 4.1



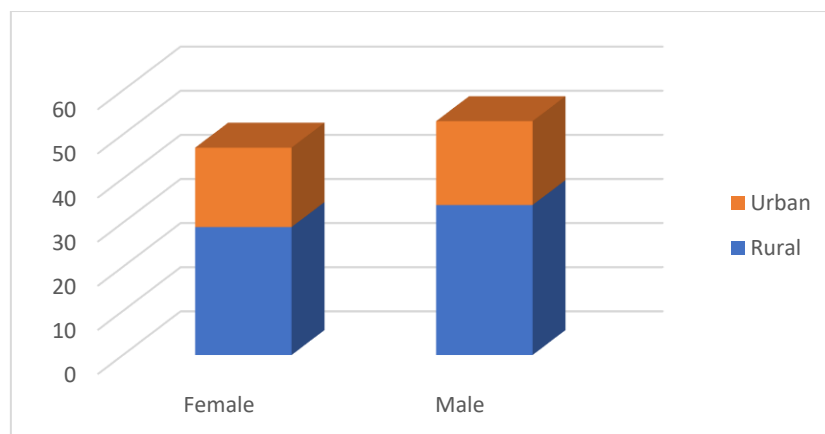
Interpretation

Female values are more stable, with smaller changes across years.

In most years, males have higher values except at year 5, where females clearly surpass males.

Table 4.2: Frequency table of gender and where do you belong.

Count of Gender	From where do you belong?		
	Gender	Rural	Urban
	Female	29	18
	Male	34	19
	Grand Total	63	37

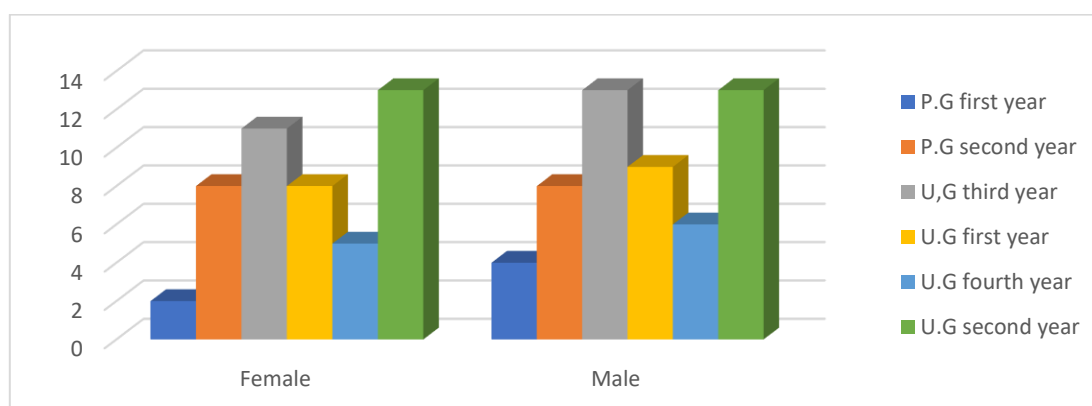
Fig 4.2: Multiple bar diagram of table 4.2

Interpretation

rural populations are larger than urban populations for both males and females, and males are more numerous than females in both settings.

Table 4.3: Frequency table of gender and education status

Count of Gender	Education status						Grand Total
	Gender	P.G First Year	P.G Second Year	U.G First Year	U.G Fourth Year	U.G Second Year	
	Female	2	8	11	8	5	13
	Male	4	8	13	9	6	13
	Grand Total	6	16	24	17	11	26

Fig 4.3: Multiple bar diagram of table 4.3**Interpretation**

Male student numbers are generally higher than female student numbers at most levels, especially:

- P.G first year
- U.G first, third, and fourth years

Table 4.4: Frequency table of gender and Room type

Count of Gender		Room type			Grand Total
Gender		Double	Single	Triple	
Female		14	15	18	47
Male		21	9	23	53
Grand Total		35	24	41	100

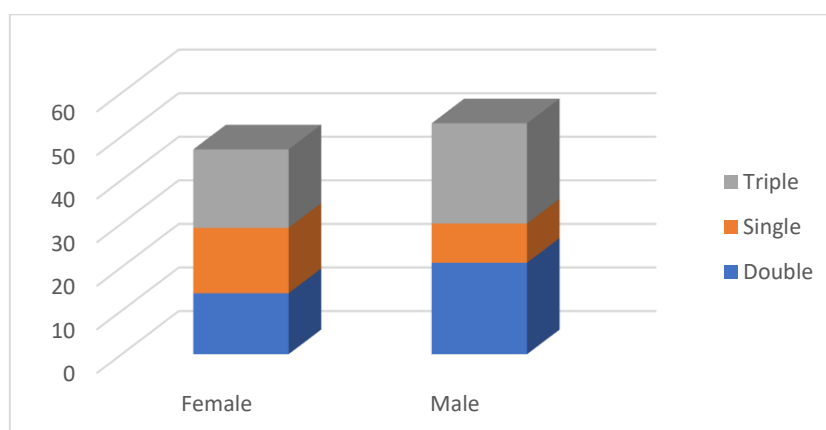
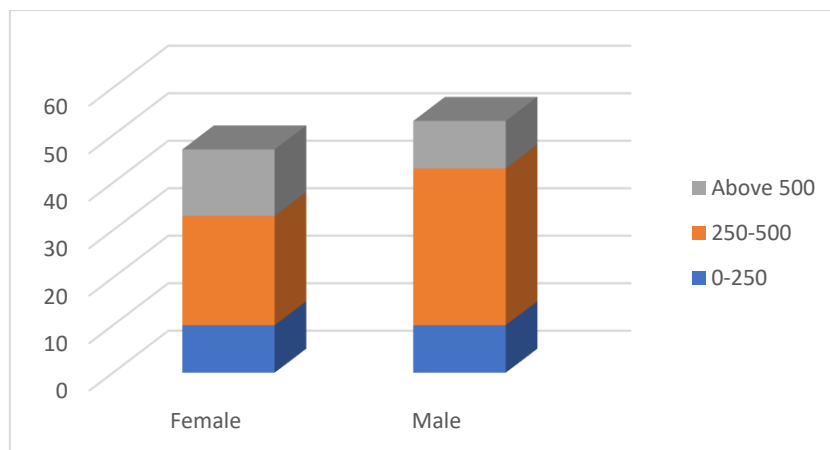
Fig 4.4: Multiple bar diagram of table 4.4

Table 4.5: Frequency table of gender and Travel expenses

Count of Gender	Travel Expenses			Grand Total
Gender	0-250	250-500	Above 500	
Female		10	23	14
Male		10	33	10
Grand Total	20	56	24	100

Fig 4.5: Multiple bar diagram of table 4.5**Table 4.6:** Frequency Table of gender and Food expenses

Count of Gender	Food expenses			Grand Total
Gender	2000-3000	3000-4000	Above 4000	
Female	15	18	14	47
Male	26	23	4	53
Grand Total	41	41	18	100

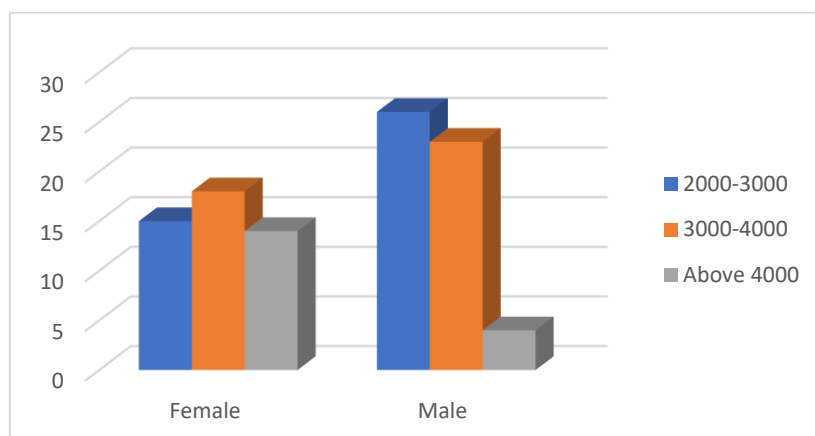
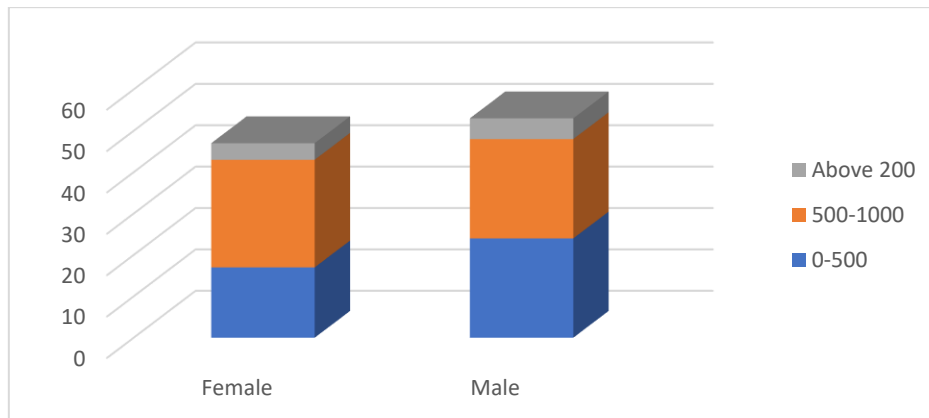
Fig 4.6: Multiple bar diagram of table 4.6

Table 4.7: Frequency Table of Gender and Entertainment

Count of Gender	Entertainment			
Gender	0-500	500-1000	Above 1000	Grand Total
Female	17	26	4	47
Male	24	24	5	53
Grand Total	41	50	9	100

Fig 4.7: Multiple bar diagram of table 4.7**Table 4.8:** Frequency Table of Gender and Study Material Expenses

Count of Gender	study material			
Gender	0-100	100-200	Above 200	Grand Total
Female	13	23	11	47
Male	17	25	11	53
Grand Total	30	48	22	100

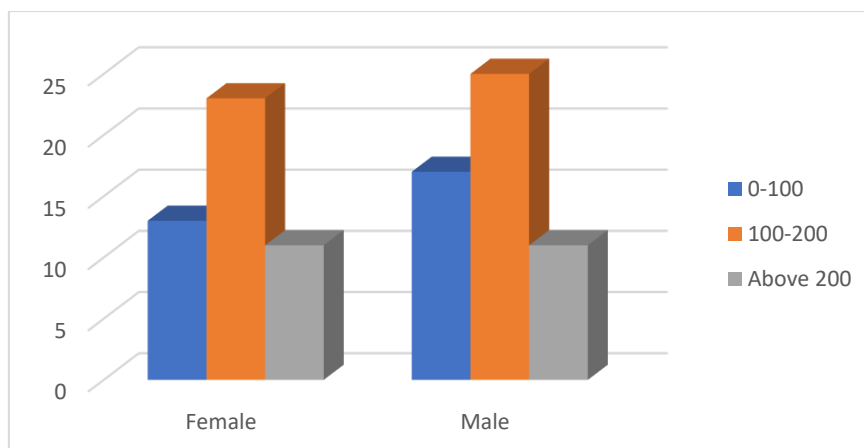
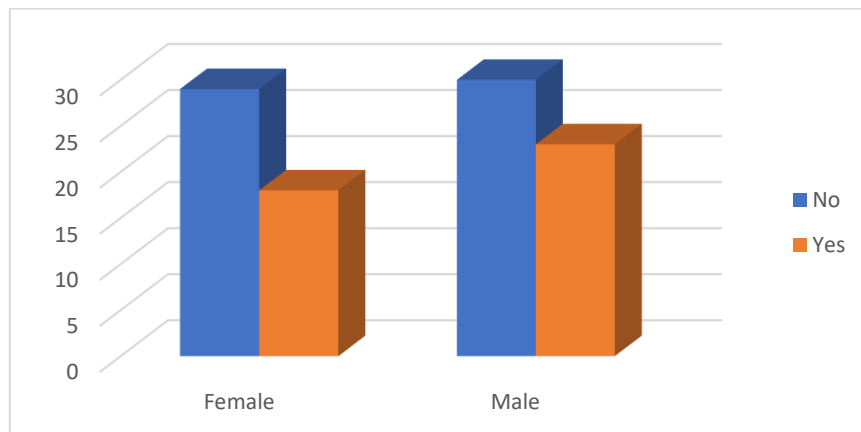
Fig 4.8: Multiple bar diagram of table 4.8

Table 4.9: Frequency table of Gender and Additional Income

Count of Gender		Additional income		Grand Total
Gender	No	Yes		
Female	29	18		47
Male	30	23		53
Grand Total	59	41		100

Fig 4.9: Multiple bar diagram of table 4.9**Table 4.10:** Frequency table of Gender and House Rent

Count of Gender		Rent			Grand Total
Gender		4500	5000	5500	
Female		19	21	7	47
Male		28	22	3	53
Grand Total		47	43	10	100

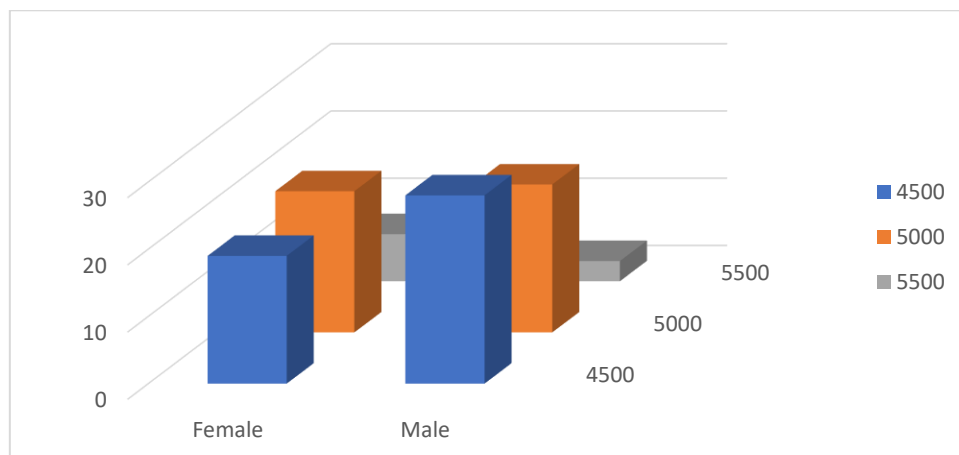
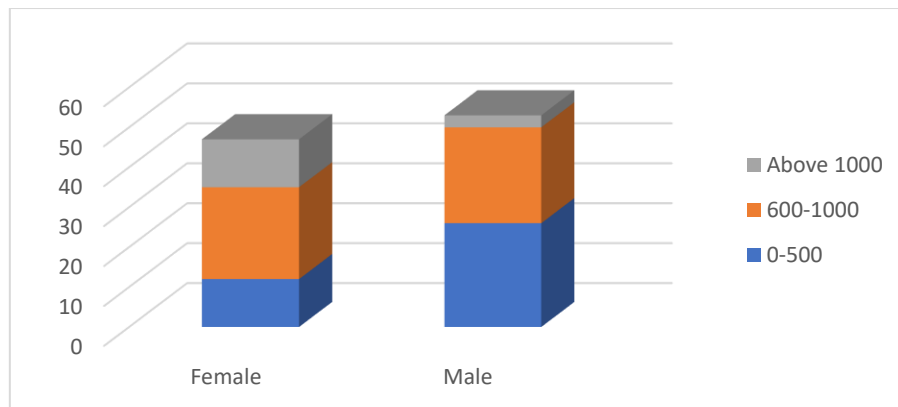
Fig 4.10: Multiple bar diagram of table 4.10

Table 4.12: Frequency table of gender and Personal/Health Care

Count of Gender	Personal And Health Care			
Row Labels	0-500	600-1000	above 1000	Grand Total
Female	12	23	13	47
Male	26	24	3	53
Grand Total	38	47	15	100

Fig 4.12: Multiple bar diagram of table 4.12**Table 4.13:** Frequency table of gender and provide wifi

Count of Gender	Provide Wifi		
Gender	No	Yes	Grand Total
Female	16	31	47
Male	19	34	53
Grand Total	35	65	100

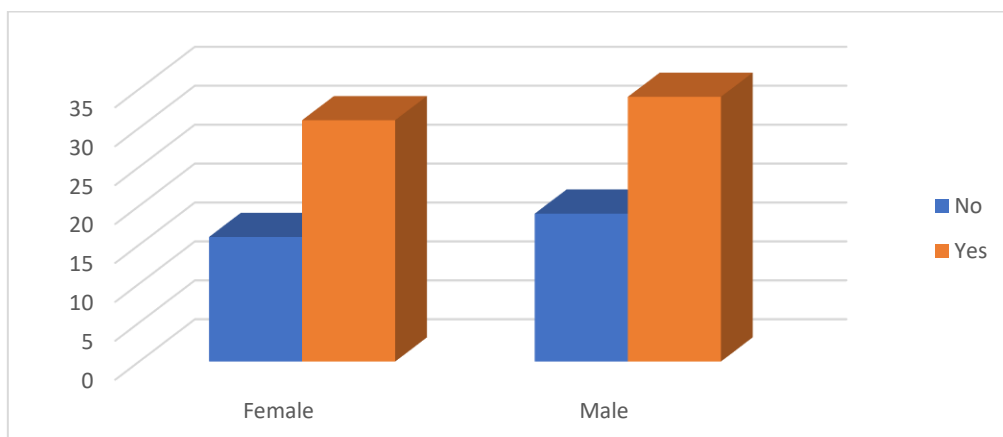
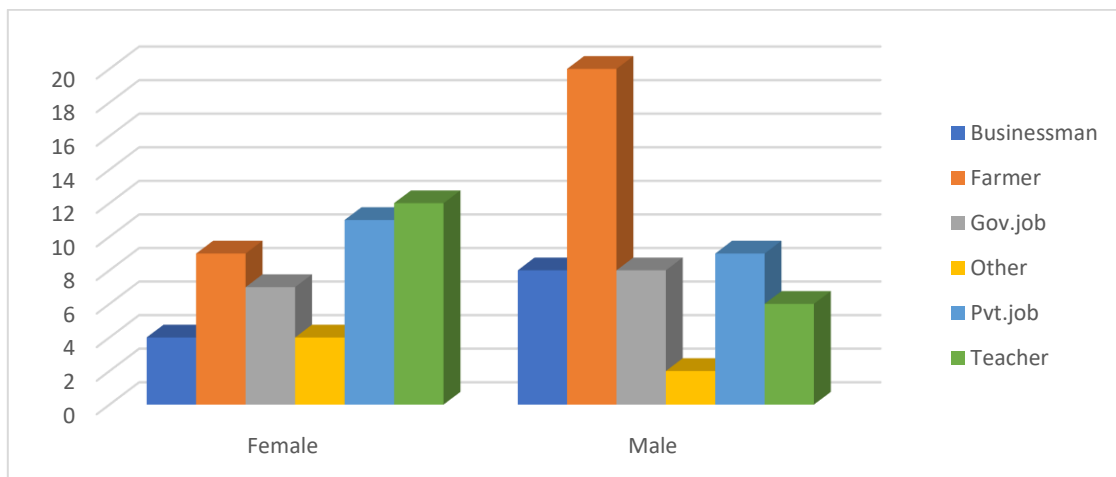
Fig 4.13: Multiple bar diagram of table 4.13

Table 4.14: Frequency table of gender and Father Occupation

Count of Gender	Father Occupation						Grand Total
Gender	Businessman	Farmer	Gov. Job	Other	Prv.job	Teacher	
Female	4	9	7	4	11	12	47
Male	8	20	8	2	9	6	53
Grand Total	12	29	15	6	20	18	100

Fig 4.14: Multiple bar diagram of table 4.14**Table 4.15:** Frequency table of gender and Mode of payments

Count of Gender	Mode of payments		Grand Total
Gender	Cash	Online	
Female	12	35	47
Male	15	38	53
Grand Total	27	73	100

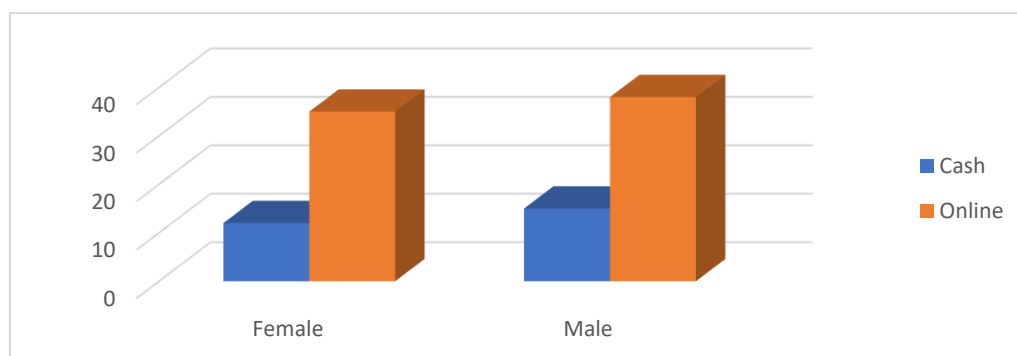
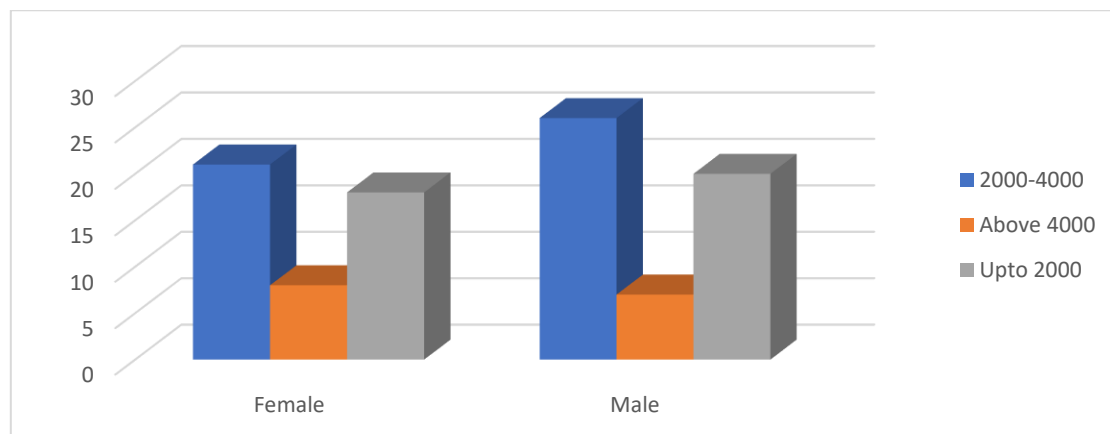
Fig 4.15: Multiple bar diagram of table 4.15

Table 4.16: Frequency table of gender and Average monthly expenses

Count of Gender	Average monthly expenses			Grand Total
Gender	Upto 2000	2000-4000	Above 4000	
Female	21	8	18	47
Male	26	7	20	53
Grand Total	47	15	38	100

Fig 4.16: Multiple bar diagram of table 4.16

4.2 Frequency Table_:

Table 4.2.1: Frequency Table of Questionnaire

Questions	Options	Frequency	% of Responce
Gender	Male	53	53
	Female	47	47
	Other	0	0
Father Occupation	Teacher	18	18
	Farmer	29	29
	Gov . job	15	15
	Pvt . job	20	20
	Businessman	12	12
	Other	6	6
	UG	78	78

Current Education Status	PG	22	22
Living Area	Urban	37	37
	Rural	63	63
Average Monthly Expenses	Upto 2000	47	47
	2000-4000	15	15
	Above 4000	38	38

4.3: Pie Charts

Fig 4.3.1:

1) what is the age ?
100 responses

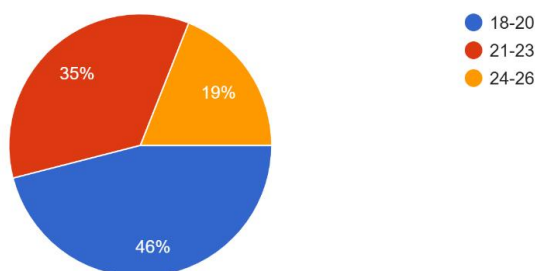


Fig 4.3.2:

2) Gender
100 responses

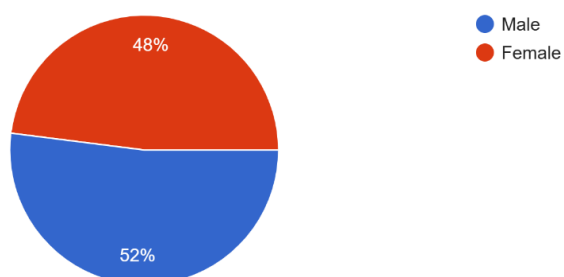
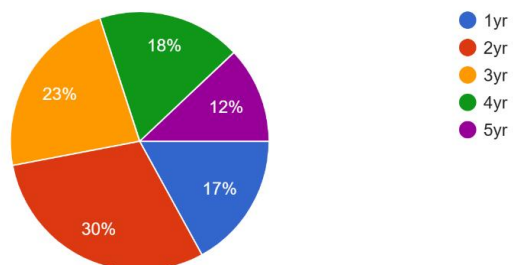


Fig 4.3.3:

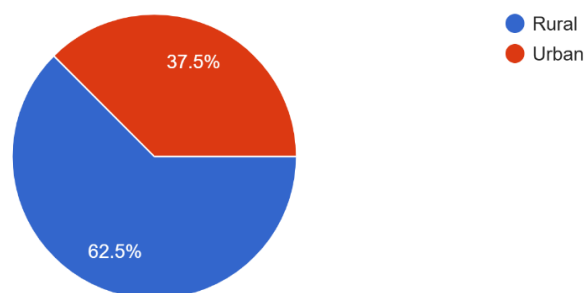
3) How long have you been living in hostel ?(years)

100 responses

**Fig 4.3.4:**

4) From where do you belong?

96 responses

**Fig 4.3.5**

5) current education status ?

100 responses

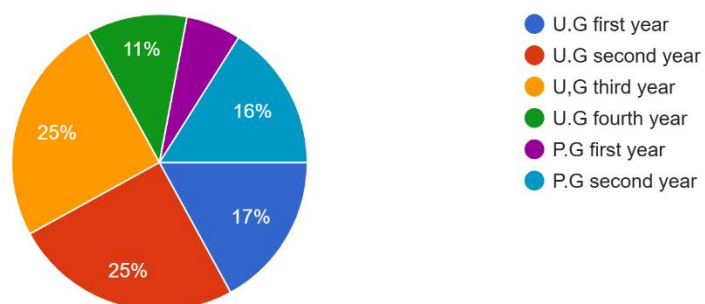
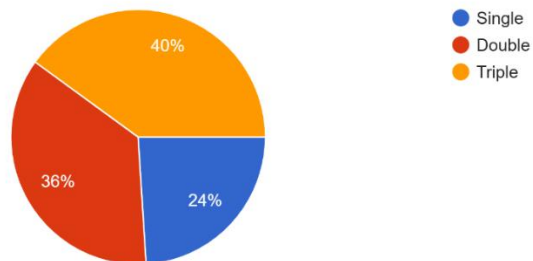


Fig 4.3.6

6) what is room are you staying in ?

100 responses



Chapter 5 :- Statistical Analysis

5.1 Chi-square Test for Independence of Two Attributes :

5.1.1 A chi-square test of independence, also known as a chi-square test of association, to determine whether two categorical variables are related. If two variables are related, the probability of one variable having a certain value is dependent on the value of the other variable. The chi-square test of independence calculations is based on the observed frequencies, which are the numbers of observations in each combined group. When we want to perform a chi-square test of independence, the best way to organize your data is a type of frequency distribution table called a contingency table. Then we have to define hypothesis based on attributes as

5.1.2 Null Hypothesis (H₀):

Variable 1 and variable 2 are not related in the population; The proportions of variable 1 are the same for different values of variable 2.

5.1.3 Alternative Hypothesis (H_a):

Variable 1 and variable 2 are related in the population; The proportions of variable 1 are not the same for different values of variable 2. Then we have to find expected values by using the following formula. If There are expected frequencies less than 5 then pooled them and make new column of them. Pearson's chi-square (X²) is the test statistic for the chi-square test of independence:

$$X^2 = \sum \frac{(O - E)^2}{E}$$

- X² is the chi-square test statistic
- Σ is the summation operator (it means "take the sum of")
- O is the observed frequency
- E is the expected frequency

5.1.4 The degrees of freedom (df):

For a chi-square test of independence, the df is $k-p-1$ where ; k – no of expected frequencies after pooled and p is the number of parameters estimated from the (sample) data used to generate the hypothesised distribution.

5.1.5 Significance level (α):

By convention, the significance level is usually 0.05. If the X^2 value is greater than the critical value, or if $p < \alpha$ then reject null hypothesis otherwise accept it.

5.2 Chi-Square Test:-

Sr. No.	Null Hypothesis	χ^2	DF	p-value	Decision	Conclusion
1	H_{01} : Gender is independent of living in hostel	15.450	4	0.004	$p\text{-value} < \alpha$, reject H_{01}	Gender is Associated With Living in Hostel.
2	H_{02} : Gender is independent of Travel spending(Per Month)	12.709	2	0.002	$p\text{-value} > \alpha$, reject H_{02}	Gender is Associated With Travel Spending
3	H_{03} : Gender is independent of Food Spending(Per Month)	17.714	2	0.001	$p\text{-value} < \alpha$, reject H_{03}	Gender is Associated With Food Spending
4	H_{04} : Gender is independent of Entertainment Spending(Per Month)	7.410	2	0.026	$p\text{-value} > \alpha$, reject H_{04}	Gender is Associated With Entertainment Spending
5	H_{05} : Gender is independent of Study Material(Per Month)	0.050	2	0.975	$p\text{-value} > \alpha$, fail to reject H_{05}	Gender is Not-Associated With Study Material

Chapter 6 :- Conclusion

6.1: Conclusion

1. Gender Differences in Spending: Males and females have different spending patterns in categories like travel, food, entertainment, and mobile recharge. For example, males spend more on travel and food, while females spend more on health care and personal expenses.
2. Living Conditions and Rent: Most students reside in triple-sharing rooms, and females generally pay more in rent compared to males.
3. Age and Education: Younger students (ages 18-20) dominate the hostel population, with undergraduates making up the majority. Expense patterns vary significantly across age and education levels.
4. Chi-Square Analysis: Statistical tests reveal that gender is associated with several factors, including spending on travel, food, entertainment, and mode of payment, but not with spending on study materials.
5. General Expense Trends: A large proportion of students, particularly females, have monthly expenses below ₹2000, while a smaller percentage spends above ₹4000. The report suggests that understanding these patterns can help administrators design better financial support systems and optimize resource allocation in hostels. For example, providing affordable Wi-Fi or subsidized transportation could address common financial challenges.

6.2: Suggestion

1. Enhanced Financial Planning Support: Offer workshops or sessions on budgeting and financial management for students to help them prioritize and efficiently manage their expenses.
2. Affordable Facilities and Amenities: Introduce or improve free WiFi services to reduce personal mobile recharge costs. Ensure affordable, nutritious meal plans in hostels to alleviate high food expenses.
3. Subsidized Transportation: Collaborate with local transportation providers to offer discounted rates for students who travel frequently.
4. Affordable Housing Options: Review rent policies to make room types, especially single and double occupancy, more affordable, ensuring they are in line with students' financial capacities.

